

SYSTEM ARCHITECTURE & HARDWARE RECOMMENDATIONS PROPOSAL

1. BUSINESS REQUIREMENTS ANALYSIS

1.1 Company Overview

Ubuntu Innovations is a fast-growing, dynamic software development and technology consulting firm. The company specializes in building modern web applications, cloud-native solutions, and custom software integrations for local and global enterprises. Grounded in the philosophy of collaborative innovation, the organization relies heavily on robust, high-availability developer environments, seamless cross-team communication, and secure handling of intellectual property (source code, proprietary client data, and development environments).

1.2 Business Goals

To maintain its competitive edge and support its fast-paced operations, Ubuntu Innovations has established several key business goals for this infrastructure overhaul:

- **Developer Productivity:** Minimize environment setup times and eliminate hardware bottlenecks that slow down code compilation and local testing.
- **Operational Continuity:** Build high-availability systems with zero single-points-of-failure to ensure developers and support staff experience uninterrupted uptime.

- **Intellectual Property Protection:** Secure proprietary source code, credentials, and client database backups against external breaches and internal leaks.
- **Cost Efficiency:** Optimize capital expenditure (CAPEX) by selecting high-performance, cost-effective hardware with long lifecycles and minimal maintenance overhead (OPEX).

1.3 Infrastructure Requirements

The underlying IT infrastructure must reliably support a hybrid team of developers, QA testers, and support personnel.

- **Compute Power:** High-core-count processors and high-capacity RAM to run local containerized environments (Docker, Kubernetes), virtual machines, and IDEs simultaneously.
- **Centralized Data Management:** Fast, localized file and asset storage with active-active backups to prevent data loss.
- **High-Speed Network Pipeline:** Minimum 1 Gbps local local-area-network (LAN) throughput with high-bandwidth Wi-Fi 6E/7 coverage to support fast transfers of large build artifacts and container images.

1.4 Security Requirements

Security is paramount to comply with industry standards (such as ISO 27001, SOC 2, and local data privacy laws).

- **Network Segmentation:** Isolated Virtual Local Area Networks (VLANs) to separate internal development, guest access, and management interfaces.

- **Unified Threat Management (UTM):** Next-Generation Firewall (NGFW) capability with Intrusion Prevention Systems (IPS), malware scanning, and content filtering.
- **Endpoint Protection:** Advanced security controls on employee machines, including full disk encryption, multi-factor authentication (MFA), and secure VPN gateways for remote access.
- **Physical & Access Security:** Centralized storage access controls, secure credential managers, and encrypted, off-site cloud backups.

1.5 Growth Requirements

Ubuntu Innovations is scaling rapidly. The selected infrastructure must accommodate this growth without requiring complete replacement.

- **Horizontal & Vertical Scalability:** Storage units must support hot-swappable drives and expansion bays. Network switches must have unused high-speed SFP+ ports for daisy-chaining or upstream fiber connections.
- **Modular Software Licensing:** Preference for open-source enterprise tools or tiered Software-as-a-Service (SaaS) options that allow user-by-user scaling without steep upfront pricing.

2. HARDWARE & SOFTWARE RECOMMENDATIONS

The following selections have been carefully curated to balance cutting-edge performance, rock-solid reliability, security compliance, and long-term cost-effectiveness.

2.1 Employee Workstations

- **Recommendation:** Lenovo ThinkPad T14s Gen 5 (AMD Ryzen 7 PRO, 32GB LPDDR5x RAM, 1TB NVMe PCIe Gen4 SSD)
- **Why Recommended:** Developers and support staff require substantial multi-tasking and compilation power. The AMD Ryzen 7 PRO processors offer industry-leading multi-threaded performance (crucial for local Docker/VM environments) paired with exceptional battery life and low thermal output. 32GB of RAM ensures that running IDEs like VS Code, multiple Docker containers, and browser tabs will not throttle system performance.
- **Business Benefits:** Minimizes compilation wait times, increases developer mobility, and reduces technical downtime. Lenovo's legendary durability ensures a 4–5 year device lifecycle.
- **Cost Effectiveness:** Considerably more cost-effective than comparable Apple Silicon alternatives or Intel-based premium laptops when factoring in standard business warranties and component upgrade paths.

2.2 Network Attached Storage (NAS)

- **Recommendation:** Synology DiskStation DS1823xs+ (8-Bay, Quad-Core AMD Ryzen, 10GbE network card installed, configured with RAID 6)

- **Why Recommended:** Codebases, media assets, database backups, and VM templates require a secure, high-speed, localized repository. An 8-bay system configured in RAID 6 protects against the simultaneous failure of two drives. The built-in 10GbE port guarantees that transferring massive files over the local network will not bottleneck daily workflows.
- **Business Benefits:** Provides decentralized, redundant backup storage that remains under corporate physical control (hybrid cloud strategy). Synology Active Backup Suite integrates seamlessly for license-free backup of all workstations.
- **Cost Effectiveness:** Eliminates the ongoing high monthly costs of storing multi-terabyte cold/warm data sets in public cloud repositories like AWS S3 or Google Cloud Storage.

2.3 Network Equipment

- **Recommendation:** Ubiquiti UniFi Ecosystem (UniFi Dream Machine SE Console, UniFi Switch Pro 24 PoE, and UniFi U6 Pro Access Points)
- **Why Recommended:** This setup provides an enterprise-grade, software-defined network (SDN). The Dream Machine SE acts as a high-performance router, firewall, and controller in one. The Switch Pro 24 provides Power-over-Ethernet (PoE) to run access points and cameras, while the U6 Pro access points ensure reliable, high-density Wi-Fi.
- **Business Benefits:** Offers a "single pane of glass" cloud dashboard for remote network monitoring, VLAN management, and rapid deployment. Built-in IPS/IDS ensures automated defense against network-level attacks.

- **Cost Effectiveness:** Zero licensing or subscription fees for the UniFi Controller platform, saving thousands of dollars annually compared to Cisco Meraki or Aruba licensing models.

2.4 Printers

- **Recommendation:** HP LaserJet Pro MFP M428fdn (Monochrome Laser Multi-function Printer)
- **Why Recommended:** In a digital-first software development environment, physical printing needs are minimal but critical when they do arise (contracts, NDAs, hardware labeling, compliance logs). This monochrome laser printer is fast, reliable, and features direct network scanning.
- **Business Benefits:** High reliability, low maintenance, and extremely fast wake-from-sleep print speeds. High-yield toner cartridges mean fewer intervention cycles for IT support.
- **Cost Effectiveness:** Laser printers have a significantly lower cost-per-page (CPP) and do not suffer from clogged ink nozzles during periods of low usage, saving on maintenance and operational overhead.

2.5 Business Software

- **Recommendation:** Google Workspace Enterprise Standard (Email, Drive, Docs, Meet) + Bitwarden Enterprise (Credential Management)
- **Why Recommended:** Google Workspace provides the gold standard for real-time collaboration on documents and presentations. Bitwarden Enterprise ensures that developers

and support staff securely generate, share, and auto-fill complex passwords and API keys, completely eliminating "password sticky notes."

- **Business Benefits:** Smooth, highly reliable productivity apps that reduce internal email overhead. Bitwarden ensures rigorous corporate password policy enforcement and credential auditing.
- **Cost Effectiveness:** Google Workspace consolidates video conferencing, email, cloud storage, and document editing into a single subscription. Bitwarden is an open-source enterprise manager that costs a fraction of competitors like 1Password or LastPass.

2.6 Development Software

- **Recommendation:** Visual Studio Code (IDE), Docker Desktop (Enterprise Edition), and GitKraken (Git Client)
- **Why Recommended:** Standardizing the core development environment ensures that onboarding is immediate and troubleshooting is uniform. VS Code is lightweight, modular, and the most widely used editor in the industry. Docker Desktop containerizes application services to prevent "it works on my machine" bugs. GitKraken simplifies complex Git branching workflows for the development team.
- **Business Benefits:** Increases developer velocity, decreases code integration friction, and ensures immediate environment synchronization.
- **Cost Effectiveness:** VS Code is entirely free and open-source. Docker Desktop and GitKraken subscriptions are

highly affordable at enterprise scale and are offset immediately by the developmental hours saved during collaborative coding.